# Markscheme

Mathematics: analysis and approaches
Practice question bank

---

**123. (a)**

Consider the word 'MATHEMATICS'.

Find the number of distinct arrangements of all the letters in this word.

there are 11 letters, with M, A and T each repeated twice *(M1)*

number of arrangements $= \dfrac{11!}{2!\,2!\,2!}$ *(M1)* **A1**

$$= \frac{39\,916\,800}{8}$$ *(M1)*

$$= 4\,989\,600$$ **A1**

**[5 marks]**